

Session 17 Quiz - AI That Combines Everything - Multimodal AI

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:57475]

Fill in the blank: In the world of AI, a 'modality' is simply a ___ of input or output.

- A. color
- B. brand
- C. speed
- D. type

Correct Answer: D. type

Explanation: The term modality refers to the specific category or "kind" of data, such as text, images, or sound, that the AI processes. It is not about color, brand, or speed.

Q2 [Type: MCQ | Difficulty: Easy | QID:57477]

Fill in the blank: The prefix 'multi' in multimodal AI tells us that the system uses ___ types of input or output at once.

- A. mathematical
- B. hidden
- C. many
- D. single

Correct Answer: C. many

Explanation: The prefix "multi" originates from Latin, meaning many or multiple. In multimodal AI, it signifies that the system combines various senses or data types (text, image, audio, etc.) at the same time.

Q3 [Type: MCQ | Difficulty: Easy | QID:57478]

Fill in the blank: Google's Veo 3 is a powerful tool designed to turn your written words into a ____.

- A. video
- B. physical robot
- C. song
- D. spreadsheet

Correct Answer: A. video

Explanation: Veo 3 is a text-to-video AI that generates moving images and often matching audio from a text prompt. It does not create robots, songs, or spreadsheets.

Q4 [Type: MCQ | Difficulty: Easy | QID:57479]

Fill in the blank: The AI tool Gamma is specifically used to transform a simple topic or outline into a full ____.

- A. social media profile
- B. slideshow
- C. printed book
- D. video game

Correct Answer: B. slideshow

Explanation: Gamma automates the creation of presentations (slideshows) by writing content and generating layouts and visuals simultaneously. It is not for social media profiles, books, or video games.

Q5 [Type: MCQ | Difficulty: Easy | QID:57480]

Fill in the blank: Whether it is a simple chatbot or a complex multimodal system, all AI follows the pattern of Sense, Think, and ____.

- A. Forget
- B. Wait
- C. Act
- D. Sleep

Correct Answer: C. Act

Explanation: The "Act" stage is the final step where the AI produces its output, such as generating text, images, or moving a robot. Forget, Wait, and Sleep are not part of this core framework.

Q6 [Type: True/False | Difficulty: Easy-Medium | QID:57481]

True or False: Multimodal AI systems are less powerful than single-modality tools because they get easily confused by different senses.

- A. False
- B. True

Correct Answer: A. False

Explanation: Multimodal AI is considered more advanced because it can process and combine multiple "senses" just like a human does, making it more powerful, not less. The statement is false.

Q7 [Type: True/False | Difficulty: Easy-Medium | QID:57483]

True or False: YouTube auto-captions use multimodal AI by listening to audio and turning it into text.

- A. True
- B. False

Correct Answer: A. True

Explanation: This involves two modalities (audio and text) working together to provide a service, which is a hallmark of multimodal AI. The AI listens to the audio and generates text captions.

Q8 [Type: True/False | Difficulty: Easy | QID:57484]

True or False: Computer Programming (Code) is considered a modality that AI can read or write.

- A. True
- B. False

Correct Answer: A. True

Explanation: The source material explicitly lists "Code" as one of the types of input or output an AI can process. Code is a text-based modality with its own syntax and structure.

Q9 [Type: True/False | Difficulty: Easy-Medium | QID:57485]

True or False: Before Multimodal AI, different AI tools (like one for text and one for voice) could easily 'talk' to each other and share ideas.

A. False

B. True

Correct Answer: A. False

Explanation: A major benefit of multimodal AI is that it brings separate abilities into one integrated system that understands how they relate. Before that, separate tools did not easily share ideas.

Q10 [Type: True/False | Difficulty: Easy-Medium | QID:57486]

True or False: A prompt like 'A dragon teaching a class' is an example of an input modality, and the generated video is the output modality.

A. False

B. True

Correct Answer: B. True

Explanation: In this scenario, text goes "in" as the prompt (input modality) and video comes "out" as the result (output modality), showing a change in modalities. The statement is correct.

Q11 [Type: MCQ | Difficulty: Medium | QID:57487]

In the AI framework 'Sense, Think, Act,' what does the 'Think' step represent for a Multimodal AI?

- A. Processing multiple types of data all at once
- B. Only looking at the words and ignoring pictures
- C. Deleting unnecessary files to save space
- D. Turning the AI's power off and on

Correct Answer: A. Processing multiple types of data all at once

Explanation: The core "magic" of Multimodal AI is its ability to understand how different senses (like vision and sound) relate to each other simultaneously during the Think step.

Q12 [Type: MCQ | Difficulty: Easy-Medium | QID:57488]

When using Google Lens to translate a foreign sign, which two modalities is the AI primarily combining?

- A. Smell and Feel
- B. Code and Music
- C. Audio and Video
- D. Image and Text

Correct Answer: D. Image and Text

Explanation: The AI uses the camera to capture an image of the sign (image modality) and then processes the characters in that image to provide translated text (text modality). It does not use smell, feel, code, music, or audio for this task.

Q13 [Type: MCQ | Difficulty: Easy-Medium | QID:57489]

Which of the following describes the 'recipe' for Voice AI as presented in the session recap?

- A. See → Process → Act
- B. Hear → Process → Speak
- C. Break → Understand → Generate
- D. Taste → Smell → Eat

Correct Answer: B. Hear → Process → Speak

Explanation: Voice AI must first detect audio input (Hear), interpret it (Process), and then generate a spoken response (Speak). See→Process→Act is for computer vision; Break→Understand→Generate is for NLP.

Q14 [Type: MCQ | Difficulty: Easy-Medium | QID:57490]

What is a major advantage of using a tool like Gamma for a school project?

- A. It can physically go to school and give the presentation for you
- B. It guarantees that your teacher will give you an A+
- C. It makes your computer run twice as fast
- D. It handles content, images, and layout in one step

Correct Answer: D. It handles content, images, and layout in one step

Explanation: Gamma's multimodal nature allows it to coordinate text writing with visual design and image selection automatically, saving time and effort. It does not go to school, guarantee grades, or speed up your computer.

Q15 [Type: MCQ | Difficulty: Medium | QID:57491]

According to the session, what is the 'careful side' or potential risk of advanced text-to-video AI?

- A. Videos take up too much space on your phone
- B. The AI might get bored of making videos
- C. It could be used to create fake videos that fool people
- D. The AI might start charging you money to watch the videos

Correct Answer: C. It could be used to create fake videos that fool people

Explanation: As AI becomes better at generating realistic videos, it becomes easier for people to create misinformation or "deepfakes." This is a major ethical and safety concern.

Q16 [Type: MCQ | Difficulty: Medium | QID:57492]

In the Veo 3 prompt example 'A cat playing piano on a rooftop at sunset,' what role does 'Motion' play in the AI's generation process?

- A. Choosing the best font for the video title
- B. Calculating the mathematical weight of the piano
- C. Deciding how the cat's paws and the light move together
- D. Checking if the cat has a library card

Correct Answer: C. Deciding how the cat's paws and the light move together

Explanation: In video generation, motion defines how the subject and environment change over time across different frames. For this prompt, motion determines the cat's paw movements and the shifting sunset light.

Q17 [Type: MCQ | Difficulty: Medium | QID:57493]

If an AI 'watches' a live soccer game and then 'writes' a summary article about it, what is the sequence of modalities?

- A. Video → Text
- B. Audio → Video
- C. Image → Code
- D. Text → Audio

Correct Answer: A. Video → Text

Explanation: The AI takes in the visual motion of the game (video modality) and outputs a written description (text modality). Image→Code would be turning a picture into code; Audio→Video and Text→Audio are different sequences.

Q18 [Type: MCQ | Difficulty: Easy | QID:57494]

Which of the following is NOT listed as a common modality in the source material?

- A. Code
- B. Gravity
- C. Video
- D. Audio

Correct Answer: B. Gravity

Explanation: Gravity is a force of physics, not a type of data input or output like text, images, code, video, or audio. The source material does not list gravity as a modality.

Q19 [Type: MCQ | Difficulty: Easy-Medium | QID:57495]

When planning a video prompt, why is it helpful to include 'Style' and 'Details' like lighting or camera angles?

- A. It makes the AI's output more specific and closer to what you imagined
- B. It helps the AI finish the video in less than one second
- C. It tells the AI which person on the internet should watch it first
- D. It makes the video file smaller and easier to send

Correct Answer: A. It makes the AI's output more specific and closer to what you imagined

Explanation: Specific descriptors help the AI narrow down the millions of possible ways to render a scene, resulting in an output that better matches your creative vision.

Q20 [Type: MCQ | Difficulty: Easy-Medium | QID:57496]

Siri uses multimodal AI because it can take a voice prompt (you talking) and provide which of the following as a result?

- A. A handwritten letter mailed to your house
- B. A physical pizza delivered to your door
- C. Only a beep and nothing else
- D. Text, images, or links on a screen

Correct Answer: D. Text, images, or links on a screen

Explanation: Siri combines the audio modality of your voice with the visual/text modalities of the search results. It1 does not mail letters, deliver pizza, or only beep.